

# TECHNICAL DESIGN

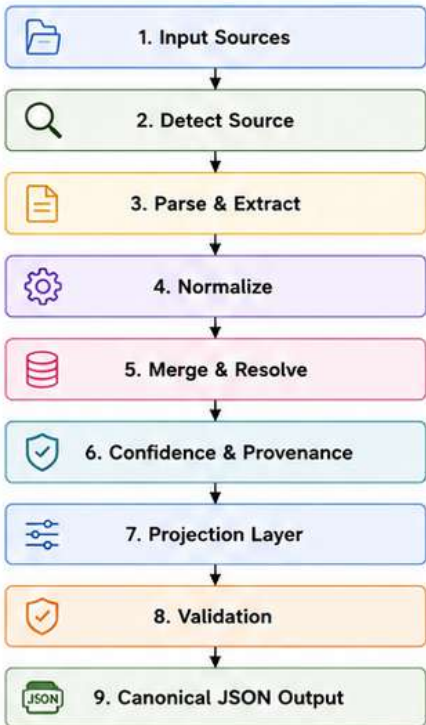
## Multi-Source Candidate Data Transformer

Author: Shreya Kumari

Email: [shreyasingh091102@gmail.com](mailto:shreyasingh091102@gmail.com)

Assignment: Eightfold Candidate Transformer

### SYSTEM ARCHITECTURE



### CANONICAL SCHEMA

|                   |                                                                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core Fields       | <ul style="list-style-type: none"><li>Candidate ID (Deterministic UUID)</li><li>Full Name</li><li>Headline</li><li>Location</li><li>Years of Experience</li></ul> |
| Collection Fields | <ul style="list-style-type: none"><li>Emails</li><li>Phones</li><li>Links</li><li>Skills</li><li>Experience</li><li>Education</li></ul>                           |
| Metadata          | <ul style="list-style-type: none"><li>Provenance</li><li>Field Confidence</li><li>Overall Confidence</li></ul>                                                    |

### NORMALIZATION STRATEGY

| Field   | Strategy                      |
|---------|-------------------------------|
| Phone   | E.164 (phonenumbers)          |
| Dates   | YYYY-MM (python-dateutil)     |
| Country | ISO-3166 Alpha-2              |
| Skills  | Canonical Mapping + RapidFuzz |

### MERGE & CONFLICT RESOLUTION

| Source Priority        | Rules                                                                                                                       |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Resume PDF (Highest)   | <ul style="list-style-type: none"><li> Resume overrides Recruiter CSV</li></ul>                                             |
| Recruiter CSV (Second) | <ul style="list-style-type: none"><li> Arrays → Union + Deduplication</li><li> Experience/Education → Fuzzy Merge</li></ul> |
| Unknown (Last)         | <ul style="list-style-type: none"><li> Winning values retain provenance</li></ul>                                           |

### EXPLAINABLE CONFIDENCE

| Source                                         | Score |
|------------------------------------------------|-------|
| Resume (Direct)                                | 0.95  |
| CSV (Direct)                                   | 0.90  |
| Regex                                          | 0.80  |
| Heuristic                                      | 0.60  |
| Unknown                                        | 0.30  |
| Overall Confidence = Average(Field Confidence) |       |

### PROVENANCE

Each field stores:

- Field
- Source
- Extraction Method
- Original Value

### VALIDATION & EDGE CASES

| Validation                                                                                                                                         | Edge Cases                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li> Canonical Schema Validation</li><li> Projection Validation</li><li> Human-readable Error Messages</li></ul> | <ul style="list-style-type: none"><li> Empty Input Files</li><li> Missing Values</li><li> Duplicate Candidates</li><li> Invalid Phone Numbers</li><li> Malformed Input</li></ul> |

### RUNTIME PROJECTION LAYER

Configured using `config.json`

Supports:

- Include / Exclude Fields
- Rename Fields
- Field Mapping
- Confidence Toggle
- Provenance Toggle
- Missing Value Behavior

Canonical Model remains unchanged.

### ASSUMPTIONS & FUTURE IMPROVEMENTS

#### Assumptions

- Same email ⇒ Same candidate
- PDFs contain searchable text
- Source files are accessible

#### Future Work

- OCR (Tesseract)
- LinkedIn Parser
- GitHub Parser
- ATS JSON/XML
- ML-based Semantic Extraction
- Parallel Batch Processing